

Derrick Lee Kuok Wei

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EDUCATION

University of Wollongong (UOW) Apr 2022 – Jun 2024
Bachelor of Computer Science (Big Data and Digital System Security)
Graduate Distinction (Highest Award)

- Applied Machine Learning libraries in classification projects for versatile model implementation.
- Experienced in ETL workflow design for cloud-native data pipelines, and SQL queries to analyse datasets.
- Proficient in team management across diverse software development projects.

PROJECT

Serverless Cloud-Native Data Pipeline (ETL Workflow) Mar 2025 – May 2025
Personal Project, Data Engineering

- Stack: Python, Terraform (IaC), AWS Services (Transfer Family, Lambda, S3, Redshift, Cloudwatch)
- Designed & built an ETL pipeline on AWS to automate data ingestion, config-driven validations, and warehousing.

Real-Time Stock Data Streaming Dashboard Feb 2025 – Mar 2025
Personal Project, Data Engineering

- Stack: Python, Python Websocket, Alpaca API, InfluxDB, AWS S3, Grafana
- Familiar with HTTP/REST APIs, WebSocket communication, and client-server architecture

Real-Time Stock Data Analysis Using Kafka Aug 2024 – Sep 2024
Personal Project, Data Engineering

- Stack: Python, Kafka, AWS Services (S3, Glue, and Athena)
- Built a real-time stock market data processing pipeline

Machine Learning Project – Credit Risk Analysis Jul 2023 – Aug 2023
School Project, Data Science

- Stack: Python, Scikit-Learn, Matplotlib, Pandas
- Achieved over 90% accuracy in predictions for potential loan defaults by training and tuning ML models.

WORK EXPERIENCE

GovTech Singapore – Data Engineering Practice Nov 2024 – Current
Data Engineer

- Authored an implementation guide translating data engineering framework into technical use for public officers.
- Built a serverless cloud-native data pipeline with an end-to-end ETL workflow, from data ingestion to warehousing.
- Co-developed a proof-of-concept RAG system to resolve log analysis pain points and streamline system audit.
- Conducted a data engineering workshop on building modern data pipeline with live demo for public officers.

National University Health System – Deputy Chief Executive Office Jun 2024 – Nov 2024
Project Manager (Data Science & Data Insights)

- Forecasted 10-year cardiac patient trends using 6 years of data with ARIMA modeling for robust predictions.
- Managed cross-institutional data initiatives to align academic research with clinical outcomes.

Future Ready Academy – Technology Education (Chinese) Jan 2024 – May 2024
Technology Educator (Generative-AI)

- Conducted workshops in NUS and NTU on robotics and gen-AI theories (NLP, LLM) in Chinese.
- Engaged foreign students with live demonstrations to support 100% tasks completion.

AWARD / CERTIFICATION

IBM Data Science Professional Certificate | Coursera Apr 2024
AWS Solution Architect Associate Certificate | Amazon Web Service May 2025

SKILLS

Languages: • Proficient in English and Chinese
Strength: • Driven Learner, Critical Thinker, Effective Communicator
Programming: • Python, SQL, Terraform, AWS Services

ACTIVITIES

Fencing Club Jun 2022 – Present
Executive Committee Member, Epeeist Fencer
Competed in a local fencing championship and secured bronze medal in men's team epee event.